Algorithm

Our project will first ask the user their name, weight and height.

Calculate the height to total inches instead of feet and inches.

Then it uses those numbers to calculate the BMI.

If the BMI is less than 30 then the program will inform the user that it is not recommended to lose weight and end the program.

If the BMI is equal or greater than 30, then we ask the user to input their gender (F or M) and their age.

If the user is older than 60 or younger than 21 then they do not meet the healthy age to fluctuate weight and the app will come to an end.

It then will ask the user to give their desired weight goal, if less than 150 then let the user know that that is below the recommended weight for the user and re-prompt the user to try again at 150 or above. We then use current weight - desired weight to get the amount of weight needed to lose. The program will then use that information to get the correct data from the database that matches the desired calories for the user's age and gender and subtract 500 calories to get the 3,500 calorie deficit goal of the week.

We then inform the user how many calories per day they need to lose 1 pound a week and how many weeks it will take to lose the desired amount of pounds. Then we will display the results in HTML (example: Jacob's weight loss goal: He weighs 175 pounds and wants to get to 150 pounds! Jacob will need to have 1800 calories per day to lose 25 pounds. We recommend having 30% Fats, 30% Proteins and 40% Carbohydrates in each meal. It will take 25 weeks to lose the weight since this goal is to lose 1 pound a week!)

Pseudocode

```
Display – "Please, enter Full Name"
        Input "FullName"
Display – "Please, enter current weight in lbs."
        Input "cweight"
Display – "Here you will be entering your height. Please, enter the number of feet:"
        Input "heightFT"
Display – "Please enter the number of inches:"
        Input "heightIN"
        Convert height entered to height in inches:
                Calculate "heightFULL": (heightFT * 12) + heightIN

Calculate BMI:
        "BMI" = 703 x cweight / heightFULL^2
        If "BMI" < 30
                Display "You are fit already, what are you doing here?"
                        Exit
                                END
        ELSE

        Display "Please, enter gender (F for Female, M for Male):"
                Input "gender"
        Display "Please, enter age:"
                Input "age"

                        If "age" <  21 or "age" > 60, then Display:
                                "You fall outside the required age range – please, consult a
                                professional for weight loss."
                                Exit
                        Else

                                UNTIL dweight > 150
                                Do
                                        Display "Enter desired weight in lbs:"
                                        Input dweight

                                        IF dweight < 150 then
```

Display "You gotta keep some meat on those bones!"
fi
DONE

Set weightToLose= “cweight” – “dweight”
Set “weeks” = “weighToLose”

The following is a table named “DailyCals” that will be built and saved in our custom database with SQLite3.

RecordKey	AgeFrom	AgeTo	Gender	Calories
1	21	40	M	2400
2	41	50	M	2200
3	51	60	M	2200
4	21	40	F	2000
5	41	50	F	1800
6	51	60	F	1600

The following is SELECT statement to calculate recommended daily calories incorporating 500 calorie deficit:
Select “Calories” – 500 as “RecCalories”
from Table “DailyCals”
where DailyCals.Gender = “gender” and
“age” between DailyCals.AgeFrom and DailyCals.AgeTo

Display
“
The following are the weight loss recommendations for *“FullName”:*
To achieve your desired weight goal of “dweight”, you must achieve a 500 calorie deficit per week for *“weeks”* weeks.

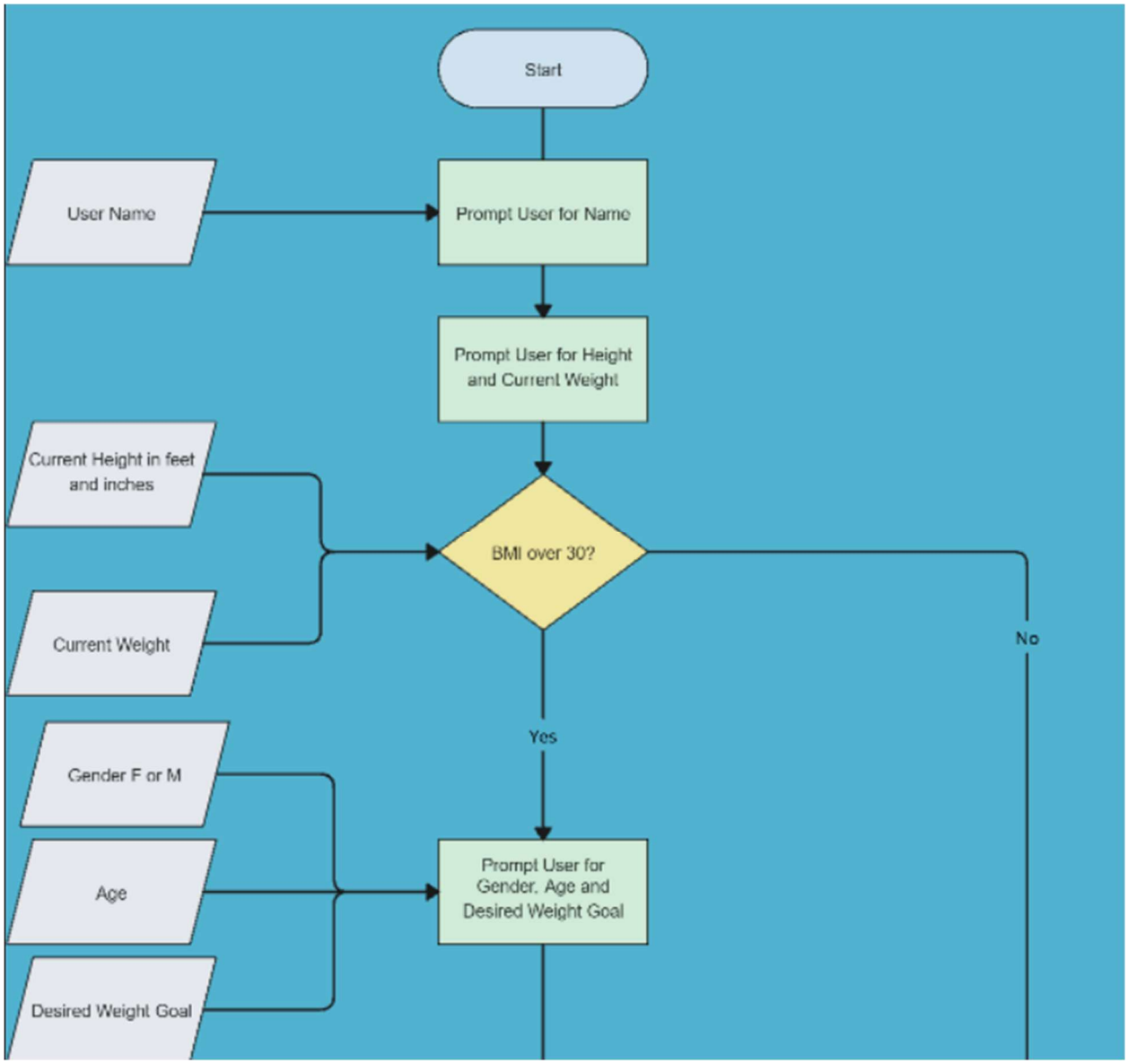
You are allotted *“RecCalories”* number of calories per day.

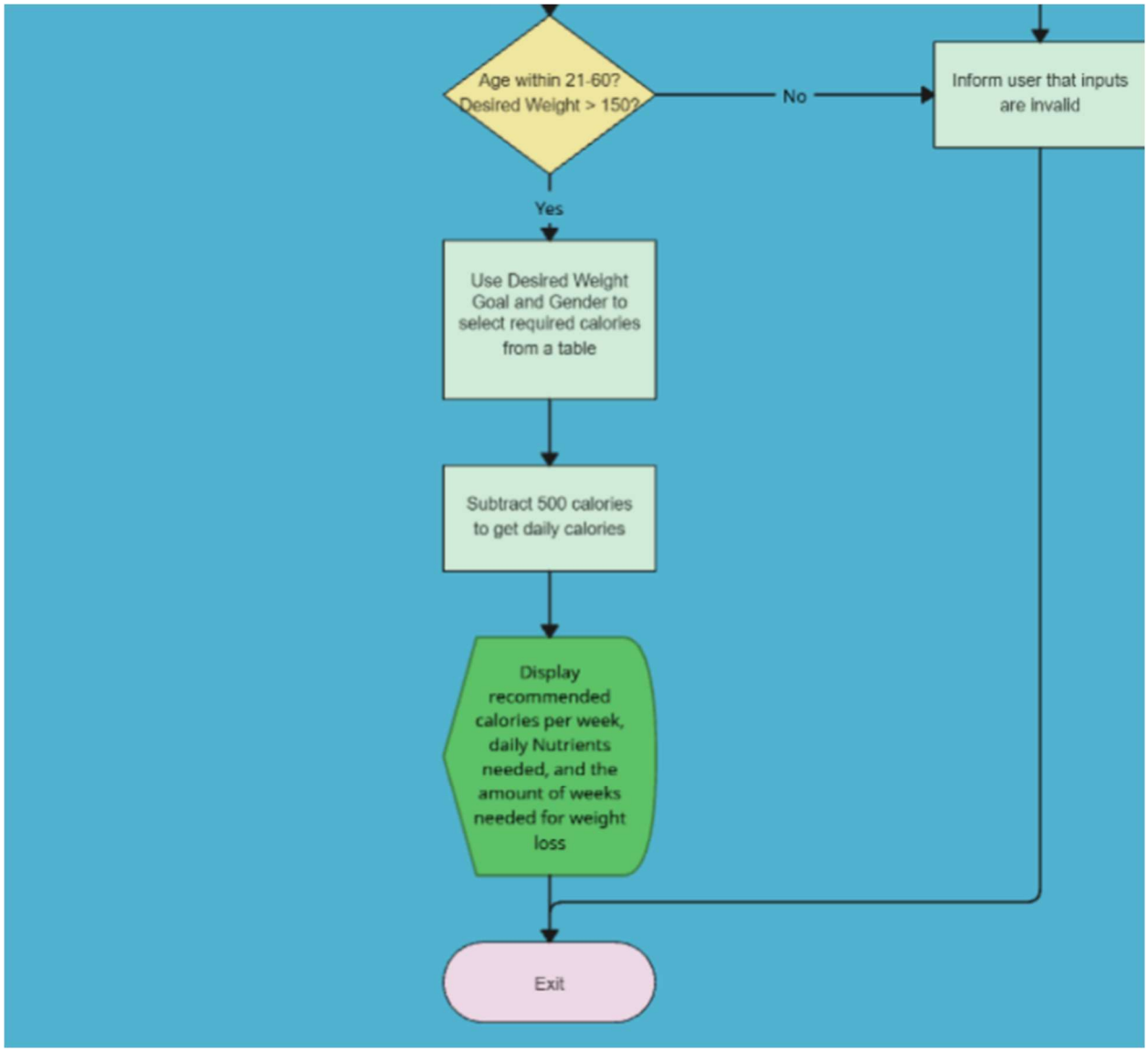
Please, refer to the following Pie Chart display of daily nutrients recommendation:

- 30% Fats
- 40% Carbohydrates
- 30% Proteins

”
Pie chart Display
END

FlowChart


Start
User Name
Prompt User for Name
Prompt User for Height and Current Weight
Current Height in feet and inches
Current Weight
BMI over 30?
No
Yes
Gender F or M
Age
Desired Weight Goal
Prompt User for Gender, Age and Desired Weight Goal


Age within 21-60?
Desired Weight > 150?
No
Inform user that inputs are invalid
Yes
Use Desired Weight Goal and Gender to select required calories from a table
Subtract 500 calories to get daily calories
Display recommended calories per week, daily Nutrients needed, and the amount of weeks needed for weight loss
Exit